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A cross-national comparison of the adoption of business process reengineering: fashion-setting networks?

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Abstract

Business Process Reengineering (BPR) was developed as a new idea which, it was suggested, could revolutionise the corporation and give it a major competitive advantage by simplifying key business processes and utilising information technology to secure cross-functional communication and focus on customer requirements. Rather than continue the debate about whether BPR is 'good' or 'bad', 'old' or 'new' (e.g. Earl, M.J., 1994. The new and the old of business process redesign. Journal of Strategic Information Systems 3 (1), 5–22; Mumford, E., 1994. New treatments or old remedies: is business process reengineering really socio-technical design. Journal of Strategic Information Systems 3 (4), 313–326), the focus here is on understanding how the concept 'BPR', however defined, has been diffused across European firms. The results from a survey of firms in four countries support previous research on the diffusion process with the rate of adoption of BPR related to micro-organisational level variables—characteristics of the firm and the boundary spanning activities of individual members. In addition there were differences across industry sectors and across countries in the extent of adoption of BPR. It is argued that attention to these meso-industry level and macro-national level factors, as well as the micro-organisational level dynamics, will improve our understanding of the relationship between networking and innovation processes. In particular, the paper focuses on how networking activity can be restrictive and lead to the creation of fashions which limit the extent of organisational learning during BPR adoption. © 1999 Elsevier Science B.V. All rights reserved.

Keywords: Business process reengineering; Networking; Innovation diffusion; Professional associations; Fashion-setting

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1. Introduction

Understanding innovation, defined as the process through which new ideas, objects and practices are created, developed, or reinvented (Van de Ven, 1986), is seen to be extremely important in the increasingly competitive and dynamic, global industrial environment. Especially in the relatively young and developing field of Information Systems (IS), innovation is extremely important, with new ideas (such as IS outsourcing, Lacity and Hirschheim (1993); decision-support systems, Keen and Scott Morton (1978); EDI, Damsgaard and Lyytinen (1997); Computer-Aided Software Engineering, Smyth (1997)) continuously entering the arena as new ‘best practice’ solutions. Understanding innovation is therefore important if those in the IS field are to influence the process whereby these new ideas are created and diffused across organisations. However, reviews suggest that ‘‘the most consistent theme found in the innovation literature is that research results have been inconsistent’’ (Wolfe, 1994: p. 405). This lack of consistency arises from the complex and contextually-sensitive nation of innovation processes. Instead, then, of looking for a single unifying theory of innovation, it is more useful to understand the conditions under which the different theories best explain particular innovation processes (Wolfe, 1994; Swanson, 1994; Damsgaard and Lyytinen, 1997). For example, Baskerville and Pries-Heje (1997), considering how well each of three different models of the innovation process explained a case study which involved the diffusion of information technology innovation within and between several commercial organisations, concluded that the models were complementary, each explaining different aspects of the case. This paper examines the diffusion of ideas about ‘Business Process Reengineering’ (BPR), a technology clearly of interest to those in the IS/IT community, and identifies a perspective which offers a good understanding of this process.

In terms of BPR itself, there are a large number of definitions and a lack of agreement about its precise nature (e.g. Hammer and Champy, 1993; Davenport, 1993; Van Ackere et al., 1993; Strassmann, 1993; Harrison and Pratt, 1993). Nevertheless, the central tenet of BPR is that organising business activities around processes rather than functions will lead to considerable increases in organisational effectiveness and so competitive advantage. Moreover, at the heart of BPR is the idea that creative use of Information Technology (IT) will provide the enabler for these changes (Fahy and O’Leary, 1997). Proponents of BPR argue that the implementation of BPR ideas involve a major departure from current practices, which therefore demands radical, rather than incremental, change in order to ensure the ‘order-of-magnitude improvements’ necessary to compete in today’s competitive markets (Davenport, 1993). BPR thus focuses on cross-functional integration, with individuals from different disciplines working together and pooling knowledge in order to solve organisational problems and exploit opportunities. In this sense BPR can be said to foster, at least potentially, what Gibbons (1995) describes as ‘‘Mode 2’’ knowledge which is multi-disciplinary, rather than ‘‘Mode 1’’ knowledge which is functionally and disciplinarily specific.

There is now increasing recognition in the literature of the limitations of BPR, even by some of its original proponents (Davenport, 1996). However, the purpose here is not to repeat debates about whether BPR is a good or bad thing (e.g. Davenport, 1996), a fad or an enduring management practice (e.g. Jackson, 1995), a new idea or an old idea repackaged (e.g. Mumford, 1994). Rather, here we are concerned with how the concept and label of ‘BPR’, however interpreted, has diffused to firms in different industrial sectors across four countries in Europe—UK, France, the Netherlands and Sweden. Examining the rate of adoption of BPR across industries and countries offers the chance of examining the wider environmental and institutional influences on this diffusion process.

In this paper we follow Damanpour and Evans (1984) in including as innovation diffusion the adoption of any product, service or process which is new to a business unit, regardless of how old (or new) it might be to another unit. Innovation thus involves organisational learning—the acquisition, sharing and utilisation of knowledge (Huber, 1991). Following this, the innovation process can be described in terms of a number of different episodes (Rogers, 1962, 1983, 1995; Clark et al., 1992). The first episode, *agenda formation*, relates to the acquisition and sharing of new ideas by organisational members. In this first episode new ideas become available within a company. The second episode, *selection*, relates to the processing of ideas within the organisation. Diverse input is gathered together and processed by multiple actors who select which particular ideas go forward for further development.

The third episode, *implementation*, describes the process of actually introducing the selected ideas as new products, services or processes to the organisation. It involves actual change, which will vary in degree from incremental (where the change simply adds to existing core competencies), through synthetic (where existing competencies are combined to form significantly new products, services or processes), to discontinuous (which involves the application of significantly new core competencies) (Tushman and Nadler, 1986). Core competencies can be described as the ‘‘collective learning of the organisation’’ (Prahalad and Hamel, 1990). BPR can be described, at least as outlined by its advocates, as a discontinuous process innovation. The final episode is *usage* and describes the situation when the innovation has become routine (Nelson and Winter, 1982). Routine invokes the notion of organisational action. The implication is that innovation becomes routine when it changes not only what a firm knows or what skills it possesses, but how it uses these. Not all new ideas that are implemented will actually be fully utilised, hence the distinction between implementation and usage. For example, Hammer and Champy (1993) themselves have suggested that 70% of BPR exercises fail.

The episodes do not represent discrete stages. The limitations of stage models of innovation, particularly in relation to complex information systems, are well known (e.g. Galliers and Sutherland, 1991; Ettlie and Bridges, 1987). Rather, these episodes are seen as iterative, overlapping and inherently political (Pfeffer, 1981; Swan and Clark, 1992), hence the term ‘episodes’ as opposed to ‘stages’. In terms of agenda formation, Adler and Shenhar (1990) identify the importance of external assets in determining an organisation’s ability to develop new products and processes. Macdonald (1995) highlights that these assets are best sought informally by individuals, rather than formally by the organisation. Individuals make contacts through a wide range of inter-organisational networks, so accessing a variety of external assets (e.g. knowledge not available internally within the firm). The individual can thus introduce new ideas into the firm (Conway, 1995). These individuals may be organisational members, who would then be described as ‘‘boundary spanners’’ (Tushman and Scanlon, 1981), introducing ideas into their firm which they have learnt about elsewhere. Alternatively, new ideas may be introduced by external individuals who may be described as ‘‘knowledge merchants’’, for example consultants and software vendors (Baxter, 1996), who are themselves constantly working to develop new ideas to introduce to clients. Without such gatekeeping activity, organisations would be deaf to outside sources of information so vital for innovation. Understanding the innovation diffusion process must therefore involve consideration of the communication networks through which individuals gain access to new ideas (Rogers, 1995).

There are a wide variety of potentially important networks, including supply chain links, links with consultants and technology vendors and links with colleagues within one's own organisation. One particular network which can be a rich source of new ideas for both boundary spanners and consultants is a professional association, the role of which is to provide its members with access to information about the latest developments in a particular knowledge domain (Lynch, 1989; Swan and Newell, 1995). Both formal knowledge transfer and personalised and informal knowledge transfer processes occur within a professional association network (Kunzel and Sadowsky, 1989). Professional associations thus influence the transparency of new ideas, making it more likely that firms will find out about others' experiences of adopting a particular new practice. Previous research has shown how professional association networks act as powerful filters in the communication of new ideas between organisations. For example, the Institute of Operations Management (IOM) has played a role in actively promoting Manufacturing Resource Planning (MRP2) technologies to manufacturing firms in the UK (Newell et al., 1996; Swan and Newell, 1995).

It is possible to relate these observations on networking to the three different theoretical perspectives proposed by Pierce and Delbecq (1977) and later Slappendel (1996), to help understand the innovation process—the individualist, structuralist and interactive perspectives. Individualist models focus on how the actions of particular people influence the development and adoption of new ideas (e.g. Rogers, 1962, 1995), hence the importance of gatekeeping and boundary spanning individuals. Research has supported the importance of individual variables in innovation processes. For example, innovators tend to be cosmopolitan (rather than local) in that they are highly involved in professional activities (Robertson and Wind, 1983; Daft, 1978) and tend to have high (rather than low) seniority (Kimberly and Evanisko, 1981; Zenger and Lawrence, 1989) and network centrality (Ibarra, 1993). These characteristics are seen to be important in that they give individuals access to ideas *and* the ability to influence others in decisions about which particular new ideas an organisation will adopt (Ibarra, 1993).

The structuralist perspective focuses at the level of the firm and seeks to find structural characteristics that influence or determine innovation, for example its size (e.g. Zaltman et al., 1973) or level of strategic planning (e.g. King, 1987). There is much evidence to support the importance of such structural variables. For example, Dobbin et al. (1988) found that larger organisations were significantly more likely to have adopted affirmative action programmes. Lawler et al. (1992) determined that firm size was positively correlated with the adoption of employee involvement practices, such as team building, self-managed teams and employee stock ownership. Johns (1993) argues that it is a combination of necessity, capacity and image which accounts for this observed relationship between innovation and organisation size. Necessity, because larger organisations are more complex and therefore more likely to be in need of regular 'fine-tuning'. Further they are more visible and so susceptible to legislative and political pressure for innovation.

Capacity for innovation comes from having slack resources, and greater likelihood of employing professional gatekeepers (such as MBAs) who can either import the latest ideas from the environment, or research them when they are suggested by senior managers (Fennel, 1984). Finally larger organisations might have a stronger desire to maintain an ‘‘image of innovativeness for external constituents’’ (Johns, 1993: p. 582). These structural factors may encourage innovation. However, the mechanisms by which these factors influence innovation are sometimes unclear. For example, larger firm size is itself associated with factors like formalisation and uniformity which mitigate against change. The structuralist perspective also tends to underemphasise the key roles played by particular individuals, such as project champions, in the innovation process (Wilson et al., 1994). Moreover, attempts to identify consistent sets of structural variables as important to innovation have failed (Wolfe, 1994).

Finally, the interactive perspective sees innovation as a result of the interaction of structural characteristics of organisations and the actions of individuals (Tushman and Nadler, 1986). For example, Nilakanta and Scamell (1990) argue that firms need to nurture in-house specialists specifically so that they can become boundary spanners and create an environment conducive to the open exchange of information. However, whether ideas introduced by boundary spanners get ‘‘put on the agenda’’ and ‘‘selected’’ will depend not only on individual characteristics of the boundary spanner (e.g. their involvement in decision-making structures, Ibarra (1993)), but also on particular characteristics of the organisation (Hage, 1980). For example, the size of the firm may influence the feasibility of implementing a particular new idea suggested by an individual. Hence, an interactive perspective is likely to provide a better explanation of this process than a focus on either individual or organisational characteristics alone. This interactive process is illustrated in a study by Robertson et al. (1996) who compared three companies of similar size and profile that had each decided to adopt an MRP2 system. In each company key individuals could be identified who introduced ideas relevant to the design of MRP2 systems to their firms. In each case individuals developed ideas about new technologies through their involvement in particular professional associations and so, in this respect, the ideas encountered were broadly similar. However, the ways in which these ideas were adopted and used in each firm depended on strongly embedded structural and cultural norms. For example, in one firm only manufacturing specialists were involved in decisions about the design of the new system. This firm had a highly differentiated functional structure such that members from other functions (e.g. purchasing) had relevant ideas to contribute but were not consulted in what was seen as a manufacturing problem. Given that MRP2 requires integration across logistics functions, the omission of a purchasing perspective proved problematic. In another firm the decision making structure was decentralised to a much greater degree and ideas introduced via purchasing specialists were taken up and integrated into the design of the new system. Thus, an interactive perspective is useful here to explain the different ways in which ideas about technological development introduced by individual boundary spanners were actually translated into organisational solutions (Schuetze, 1996).

All three of these perspectives, however, focus at a relatively micro-level of analysis, that is at the level of the individual and the organisation. More recently it has been recognised that meso- and macro-level factors need to be taken into account in understanding the diffusion process as well as the micro intra-organisational level factors (Clark and Newell, 1993). As Damsgaard and Lyytinen (1997) state, the micro-level of analysis ‘‘cannot account for differences in diffusion patterns due to variances in environmental and institutional factors’’ (p. 43). Research on the innovation diffusion process needs, therefore, to emphasise the context-dependent nature of the innovation process. This context dependency is related to the highly social nature of the innovation process.

The macro-level social context of an organisation constitutes a large number of broad elements such as the nature of the legislative system within which an organisation operates, the national culture, national institutions such as professional or trade associations, or the character of labour relations at the supra-organisational level (Clark, 1987; Bessant and Grunt, 1985). For example, national professional associations have been found to play a role in the promotion of MRP2 as ‘best practice’ in the UK and North America (Swan and Clark, 1992). An organisation’s national government is one of the most significant elements of the macro-context. For example, in Germany, the higher level of trade union involvement in organisational decision-making compared to many other countries including the UK, has arguably been a factor in limiting the rate at which German companies have adopted the type of flexible employment and work patterns which became widely used in the UK throughout the 1980s.

The meso-level context of an organisation constitutes the level intermediate to the macro-level of government policy and the micro-level of intra-organisational dynamics. For example, Spender (1989) refers to ‘‘industry-level recipes’’, which are said to emerge as unintended consequences of managers’ need to communicate with others within their industry because of the inherent uncertainties of organisational life. These industry-level recipes refer to shared patterns of managerial judgements concerning issues of product, technology, marketing, personnel etc. Following Sorensen and Levold (1992) the meso-level is seen primarily to concern the institutional dynamics of inter-organisational *network* relations, which as outlined, is a central part of the innovation process. Research on both the diffusion and utilisation of new technologies shows the importance of the meso-level context in shaping the character of the implementation process. For example, in a study of the financial services sector Scarbrough (1996b) illustrated how the implementation of new IT systems was influenced by the ways each bank attempted to manage their network relations and control the technologies and knowledge developed.

This paper looks at factors at all three levels (macro-level: nationality; meso-level: industry sector; and micro-level: organisational size and strategic planning and individual networking activity) in terms of their usefulness in explaining the diffusion of BPR.

2. Method

The study presented here is part of a wider research project[1] which has involved analysing the role of networks in the diffusion of information technologies which support

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operations management. This project has focused in particular on the role of professional association networks in diffusing ideas about operations management which may then influence the adoption of information technologies by firms in different sectors across 4 countries in Europe—UK, France, the Netherlands and Sweden. Individuals who are involved in such associations potentially act as boundary spanners, learning about the latest developments in their field. Members of the professional associations most closely aligned to the knowledge domain of Operations Management in each country have been surveyed.[2] These professional associations are all affiliated to the American Production and Inventory Control Society (APICS), so that there is a common base in their focus and in the material they diffuse. The main source of data was obtained from a survey of these professional association members. Questionnaires were translated into the appropriate language and piloted on individuals familiar with the technologies, as well as the particular language to ensure that the technology 'jargon' was accurate. In addition, interviews were conducted with key boundary spanners (i.e. practitioners, consultants and software vendors) in each country (approximately 20 interviews in each of the 4 countries) and this qualitative data is used to aid the interpretation of the quantitative data from the surveys.

This methodology provides information about a firm from the perspective of one organisational employee only. This inevitably produces some bias, as different employees might give somewhat different answers to questions about their firm. It also provides information on the boundary spanning activity of only one employee. These are limitations of the methodology adopted, but these need to be weighed against the advantages of being able to make comparisons across a large sample of firms in different industries and different countries.

An indicator of diffusion is to look at the spread of adoption of a particular technology. The survey asked respondents to indicate from a list of technologies those that had been adopted in their own organisation (e.g. TQM, MRP2, JIT). For each technology that had been adopted, respondents indicated the length of time since its adoption (on a scale from 1 = not adopted; to 4 = adopted for more than 3 years). Included in this list was BPR (defined simply as ''Business Process Reengineering'' or its equivalent translation). Data is included in the analysis below where respondents provided information on this 'BPR' question (n = 1277). Questions were also asked about variables that the literature suggests would influence the rate of diffusion of such a new idea. The variables included were not exhaustive but comprised variables from the three different levels of analysis identified as important in the literature on innovation diffusion. These were:

1. Micro-Organisational Level

   Individual Activity-the networking activity of the individual including their involvement in the professional association activities.
   Organisational Characteristics-which included both structural variables (firm size) and process variables (technology strategy).

2. Meso-Industry Level

   Industry sector—the type of industry.

[2] The Professional Associations involved in the study were: in the UK the Institute of Operations Management, IOM; in Sweden PLAN; in the Netherlands VLM; and in France CPIM de France.

3. Macro-National Level

Country—UK, France, the Netherlands and Sweden.

This quantitative data is used here to establish which variables differentiate between firms that had adopted BPR and firms which had not adopted BPR. The qualitative data was useful for explaining some of these differences.

3. Results

Data is included in the analysis where respondents provided information about whether BPR had been adopted in their firms (UK n = 733, France n = 170, Netherlands n = 198, Sweden n = 176). Of the total sample for which data about the adoption of BPR was provided, 71% (n = 912) indicated that their firms had not adopted BPR and 29% (n = 365) that their firms had adopted BPR, the majority of these firms having adopted within the last 2 years. This is to be expected given that BPR is a relatively new idea. In the analyses below, independent sample t-tests and chi-squares analyses are used as appropriate to compare firms where BPR had been adopted, with firms where it had not been adopted. In addition, regression analysis was used to consider the relative strength of the different variables in predicting the rate of BPR adoption.

3.1. Adoption of BPR

3.1.1. Micro-organisational level variables

3.1.1.1. Individual networking. Individuals were asked how often they used various networks to find out about the latest ideas in their particular field of operations management. They were asked to respond, in relation to each of 10 different networks, on a 6 point scale from never (1) to daily (6). Four distinct types of networks were identified and the mean score was computed for each. These were *colleague networks*, which included contact with colleagues within the respondent's own department, colleagues in other departments and colleagues in the wider corporation; *professional*

Table 1
Mean networking score for adopters and non-adopters on the various networks used to find out about the latest ideas

	Non-Adopters[a]	Adopters[a]	T-value
Colleague networks	4.19	4.35	2.53[b]
Professional networks	2.27	2.18	NS
Supply chain networks	3.39	3.39	NS
Consultant/vendor networks	2.03	2.26	2.25[c]
Total contact with networks	2.65	2.72	NS

[a] 1 = 'never' to 6 = 'daily'
[b] Significant at the 0.05 level
[c] Significant at the 0.01 level

networks, which included contact with members of the focal professional association (e.g. other IOM members for UK respondents), members of other professional associations and members of professional user-groups to which the person might belong; *supply chain networks*, which included contact with suppliers and customers; and *consultant/vendor networks*, which included contact with consultants and hardware and software vendors.

As can be seen from Table 1, individuals in firms where BPR had been adopted were significantly more actively involved in networking with colleagues within their own organisation. Individuals in firms where BPR had been adopted also had significantly more contact with consultant/vendor networks. Again, this reflects the fact that individuals in firms which had adopted BPR had more contact than individuals in firms which had not adopted BPR with both software and hardware vendors and with consultants. There were no differences between the two groups in terms of professional networking or supply-chain networking. When the average level of contact across all the networks was computed the results showed that there was no significant difference between individuals in firms which had adopted and individuals in firms which had not adopted BPR.

3.1.1.2. Individual involvement in professional association activities. Individuals were asked a number of questions about their involvement in the focal operations management professional association (the specific association being unique to the respondents in each country). They were asked how often they had attended the various events sponsored by the association in the last 12 months (for each type of event they were asked to respond on a 3 point scale from 1 = never attended in the last 12 months; 2 = attended one such event in the last 12 months; 3 = attended more than one such event in the last 12 months). The events included both formal events—seminars, annual conferences, courses and meetings (formal events); and events geared to more informal discussion and observation—factory visits and social events (informal events). These events were common across the four countries with the exception of France which did not organise seminars. Attendance at these various events was not particularly high. Factory visits were the most frequently attended event, although even here 77% had not attended such an event in the last year. Computing mean attendance scores, Table 2 shows that individuals in firms where BPR had been adopted were significantly more active in their attendance at the professional association events. Analysis of formal and informal events separately indicated that this higher level of attendance was true for both the formal and informal events organised by the respective associations.

Table 2
Mean score for adopters and non-adopters on attendance at professional association events

	Non-adopters	Adopters	T-Value
Mean total events	1.22	1.29	3.17[a]
Mean formal events	1.23	1.31	3.12[a]
Mean informal events	1.19	1.27	2.75[a]

[a] Significant at the 0.001 level

Table 3
Difference between adopters and non-adopters of BPR and organisation size

	Non-adopters	Adopters
Less than 100	193 (78%)	54 (22%)
100–500	381 (80%)	98 (21%)
501–1000	146 (66%)	77 (35%)
1000 +	188 (58%)	135 (42%)

3.1.1.3. Organisational size. Respondents were asked to indicate the size (as measured in terms of the number of employees) of their own organisation on a 7 point scale (from 1 = less than 50 to 7 = more than 5000). Grouping some of these categories together, Table 3 shows that there was a direct relationship between firm size and adoption of BPR.

Table 3 shows that the firms where BPR had been adopted were significantly larger than the firms where BPR had not been adopted ($t = 6.38$, $p < 0.0001$).

3.1.1.4. Technology strategy. Respondents were asked three questions concerning their perceptions of their firms' technology strategy—whether their firm had developed a technology strategy, whether they felt this technology strategy was clearly defined and how far the strategy was linked to the business strategy of their firm—each of which they had to answer using a 5-point scale, (from 1 = very untrue to 5 = very true).[3] Comparing individuals in firms which had and had not adopted BPR, where BPR had been adopted the individual was more likely to agree that their firm had developed a technology strategy ($t = 4.6$, $p < 0.0001$), that this strategy was clearly defined ($t = 5.38$, $p < 0.0001$) and that it was linked to the business strategy ($t = 5.45$, $p < 0.0001$).

3.1.2. Meso-industry level

3.1.2.1. Industry sector. The majority of survey respondents worked in the manufacturing industry, which was expected given the focus of the particular professional associations. However, other industry sectors were represented by the survey respondents. There were significant differences (Chi = 26.61, df = 7, $p = 0.00039$) between industry sectors in the proportion of firms that had adopted BPR, as seen in Table 4.

Interestingly, consultancy firms appeared as the most likely type of organisation to have adopted BPR, but most of the other sectors (the exception being education) had a higher rate of BPR adoption than the manufacturing sector. Perhaps the more complex structure of a manufacturing firm, making integration across traditional functional departments more problematic, accounts for this lower rate of adoption. The much higher adoption rate among those 'selling' the idea of BPR—consultants and vendors—suggests that they

[3]These questions are relatively simplistic, but broadly map on to the distinction between 'information strategy'—whether there was a technology strategy; 'IT strategy'—how far this was clearly defined; and 'IM strategy'—how far the technology strategy was linked to the business strategy (see Earl, 1989).

Table 4
Difference between adopters and non-adopters of BPR and industry sector

	Non-adopters	Adopters
Manufacturing	710 (75%)	243 (25%)
Service/retail	38 (62%)	23 (38%)
Transport/distribution	26 (67%)	13 (33%)
Software vendors	28 (66%)	25 (34%)
Hardware vendors	10 (59%)	7 (41%)
Education	15 (88%)	2 (12%)
Consultancy	49 (56%)	39 (44%)
Other	16 (55%)	13 (45%)

may be less aware of the difficulties which those in the more complex manufacturing environment may face.

3.1.3. Macro-national variables

3.1.3.1. Country differences. There was a significant difference (Chi = 33.54, df = 3, $p < 0.0001$) across the four countries in the rate of adoption of BPR, as seen in Table 5. Table 5 shows that significantly fewer firms in the UK and Sweden had adopted BPR compared to France and the Netherlands.

3.1.4. Regression analysis

Given the range of variables which were found to be related to the adoption of BPR, a regression analysis was conducted to assess the relative strength of these variables. The stepwise method of inclusion was used with the level of significance for inclusion set at 0.05. The dependent variable was the adoption of BPR with scores ranging from 1 where BPR had not been adopted through to 4 where it had been adopted for more than 3 years. All the variables found in the previous analysis to be significantly related to adoption of BPR were included (extent of networking with colleagues and with consultants and vendors; extent of involvement in professional association activities both formal and informal; extent of technology strategy definition; organisation size; industry sector; and country). Table 6 shows the variables that significantly predicted adoption in the order in which they were entered into the equation.

The amount of variance explained is small (Total R square = 0.08), but this is to be expected given that the activity of only one individual is included in the analysis. More importantly it demonstrates that a combination of organisational, industry and national variables are needed to explain the variance in BPR adoption.

Table 5
Percentage of firms adopting BPR in the four countries

UK	France	Netherlands	Sweden
168 (23%)	63 (37%)	82 (41%)	52 (30%)

4. Discussion and conclusions

These findings support previous research concerning the micro-level organisational factors influencing the adoption of BPR. They highlight the importance of organisational size (e.g. Preece and Edwards, 1993) and, in particular, the importance of IT and business strategy in BPR adoption (Ettlie and Bridges, 1987; King, 1987; Galliers, 1993). As Paolucci et al. (1997) (p. 588) state: ''Process innovation is very difficult unless the analysis is focused on the entire business of a firm (or on a part of it), with particular strategic objectives in mind''. The networking activity of the individual members was also a significant micro-level factor. In particular, the results showed that individuals in firms which had adopted BPR had more contact with consultants and vendors and more involvement in the focal professional association activities than did individuals in firms which had not adopted BPR. Of course, from the particular methodology used in this study it is not possible to establish whether this greater networking activity influenced the BPR adoption in the individual's own firm or whether BPR adoption led to the increased activity in these networks. However, it is clear that the more contact an individual has with these types of networks the more likely they are to be made aware of the latest, 'best-practice' ideas, which in this case is BPR.

This study also suggests that micro-level factors alone are not sufficient to understand the adoption of BPR. Micro processes such as individual boundary spanning activity will be shaped by the wider meso- and macro-level contexts in which these processes occur. Thus, there were significant differences between industry sectors and between the four countries in the extent of BPR adoption. This in part can be explained by the differences across industry sectors and countries in the institutions which support networking between individuals and firms. For example, interview data indicated that the professional associations in each of the four countries operated differently in terms of both the way they formally promoted new ideas such as BPR and in the way they facilitated informal networking between their members. Moreover, consultants had a rather different role across the four countries and this influenced the ways in which they promoted the diffusion of particular ''best practice'' ideas (Robertson et al., 1997). There were also national differences in the way the concept 'BPR' was being used that was evident from the interviews. For example, interviewees in Sweden talked quite a lot about process management but were reluctant to call it 'BPR' because they tended to see it as an incremental, rather than a revolutionary process. What is required is an extension to the focus on

Table 6
Regression analysis with length of BPR adoption as the dependent variable

	F value	R	R square
Size	21.92	0.15	0.02
Industry sector	29.04	0.23	0.05
Country	23.04	0.25	0.06
Strategy	19.42	0.27	0.07
Involvement in informal PA events	16.72	0.28	0.08
Consultant/vendor networking	14.85	0.29	0.08

micro-level factors which have dominated the literature to date to include the more meso, industry-level factors and macro, national-level institutional factors as well. One particular perspective which focuses on these more macro-level influences on diffusion is Abrahamson's analysis of management fashions.

Abrahamson (1991) contrasts theoretical approaches which emphasise on the one hand imitation and legitimacy and on the other hand those that emphasise efficiency criteria as the primary explanation for choices about technology adoption. He argues that imitation is more important in explaining the broad diffusion of new ideas and describes this as a fashion-making process, with fashion defined as ''a relatively transitory collective belief, disseminated by management fashion setters, that a management technique leads to rational management progress'' (Abrahamson, 1991: p. 257). He argued that management fashions are generated because of the cultural value placed on norms of rationality and progress. This means that managers will seek to create an image of rationality by adopting technologies that gain legitimacy in their field as new forms of 'best practice'. Moreover, the extent to which norms of rationality and progress guide choices about the adoption of technologies varies across countries (Abrahamson, 1996). This study lends support to the existence of country differences in diffusion patterns (see also Clark and Newell (1993)).

Blumer (1969) noted that new administrative models became fashionable through the active role of ''fashion setters'', who select particular models and then spread awareness of these. While it is possible that the initial selection is based on technical efficiency criteria, it is also at least possible that marketability will be an important influence. For example, Johns (1993) concludes from his research that the technical merits of an idea account for little of the variance in its adoption. In an interesting study of the diffusion of quality circles in the USA, Ledford et al. (1988) contrasted the continued popularity of this innovation with the documentation of the frequent failure of quality circles. Based on their findings Johns (1993) argues that the diffusion and adoption of quality circles was based less on technical merit than on the symbolic, cultural properties of the choice. He argues that the diffusion of quality circles was strongly encouraged by the emergence of an international quality circle association, a journal, and the registration of certified consultants. He concludes that ''These political and institutional infrastructures surely cloud natural selection.'' (Johns, 1993: p. 581). The importance of these meso- and macro-level contextual factors was supported in this current study of the diffusion of BPR. Atewell (1992) similarly asserts that supply-side institutions affect adoption patterns as much as ''true demand'' by adopters.

Professional associations may be a particularly important supply-side institution in this fashion-spreading process because they bring together both technology suppliers (software vendors and consultants) and users, thus providing a forum for discussing the latest ideas. Moreover, information use is influenced by the users' perceptions or attitudes about the source of information (Cullen, 1983) and a professional association tends to be perceived as credible, based on ''professional values''. However, in previous work Newell et al. (1996) noted how professional association dissemination activity is dominated by consultants and software vendors who use the association to market their products and services. Thus, although it may be through the activity of individual boundary spanners that new ideas get placed on an organisation's agenda, what these individuals 'find' is partial and restricted. Suppliers tend to identify and promote a (singular) best practice innovation that can be slotted into an organisation in the (singular) most effective way (e.g. Scott-Morton, 1991). As Sanchez (1991) notes, the efforts made by suppliers to inform potential adopters of their products have a positive effect on the diffusion rate of new technology.

Describing BPR as a fashion suggests that ‘‘fashion setters’’, including consultants and software vendors, have chosen this particular model as exploitable, emphasising its increased rationality over existing ways of organising. This is supported here where firms that had adopted BPR had significantly more links with consultants and software vendors than those who had not adopted BPR. Even though it is well known that there is a high failure rate associated with BPR, it is still marketed as a rational way of managing on the basis of the dramatic improvements associated with the 30% of firms which do succeed—a sort of ‘‘red badge of courage’’ (Fahy and O’Leary, 1997).

Abrahamson (1991) argues that even if fads and fashions are inefficient, it cannot be concluded that they necessarily harm organisations: ‘‘Even if quality circles did not enhance the performance of a single US organisation, they may have refocused attention on the notion of quality and worker participation’’ (Abrahamson, 1991: p. 608). Likewise, other fads and fashions may be important in drawing attention to problems and solutions that have been previously overlooked. However, taking a different perspective, Tushman and Nadler (1986) describe the most innovative organisations as highly effective *learning systems*, acquiring or gathering diverse information about customers, competitors and technologies *and* involving multiple actors in processing it in order to solve problems. When an idea is adopted because it is a ‘fashion’, it suggests that the problem-solving or selection episode of the innovation process has been short-circuited. For example, Loh and Venkatraman (1992) consider how Eastman Kodak’s success in outsourcing its IS department triggered a ‘‘bandwagon’’ effect, with many firms copying Kodak’s strategy in the hope of gaining the same success. They argue that a number of ‘‘myths’’ have developed around why IS outsourcing is more efficient which encourage firms to copy this strategy without a full analysis of the costs and benefits. Their own research demonstrates the falsity of these myths which means that outsourcing is not always beneficial. Similarly, Scarbrough (1996), discussing the diffusion of BPR, suggested that managers copy the approaches developed by their rivals because they fear the consequences of not doing so, rather than because they have systematically assessed the advantages of adoption. The idea has been acquired through the boundary-spanning activity of organisational members, put on the agenda and simply implemented because others are using it. This could be described as a process of mimetic isomorphism (DiMaggio and Powell, 1983).

The high degree of reported failure of BPR can, then, be linked to the way BPR has been diffused as the ‘latest fashion’. The adoption of BPR, as described by its supporters (e.g. Hammer and Champy, 1993), will lead to the exploration of new opportunities and alternatives. Indeed, BPR may be particularly important in this respect since, as March and Simon (1958) note, departmental structures can inhibit learning, by focusing their members on parochial rather than organisation-wide problems. BPR, by breaking down these barriers, can potentially foster organisation-wide learning. However, BPR as a fashion will merely, at best, exploit existing competencies, and at worst undermine and not replace existing competencies. Thus, if innovation is related to organisational learning it must involve the dual processes of knowledge acquisition and knowledge processing (Huber, 1991). Technologies become effective only through gradual, careful, and sustained implementation processes that provide organisations with the tacit knowledge and the skills necessary to implement these technologies efficiently. In terms of knowledge acquisition the problem is that what is acquired through the activities of individual boundary spanners is potentially partial and biased. Partial because, coming via a professional association, it is disciplinary-based (mode 1 knowledge) rather than multi-disciplinary (mode 2 knowledge) (Gibbons, 1995) and because knowledge merchants have a vested interest in marketing only certain types of knowledge. Biased because what are shared (e.g. between knowledge merchants and users) are typically stories of success, while stories of failure, at least in the public domain, are rare. In terms of knowledge processing, the problem is that an idea is often adopted through imitation processes, rather than problem-solving processes.

Describing BPR as a ‘fashion’ suggests that its diffusion is explained less as an attempt to optimise actions than as motivated by political concerns (Cooper and Zmud, 1990). Davenport (1996) himself (one of the early BPR gurus) raises the question ‘‘How did a modest insight become the world’s leading management fad?’’ He argues that it was because the concept of reengineering brought together an ‘‘iron triangle’’ of powerful interest groups—top managers at big companies, the big management consultants and the big information technology vendors. Managers rewrote their histories so that past examples of successes became repackaged as reengineering success stories. At the same time as this repackaging of projects was occurring, consulting firms were repackaging their expertise: Continuous improvement, systems analysis, industrial engineering, cycle time reduction—they all became versions of reengineering. A feeding frenzy was under way. Major consulting firms could routinely bill clients at US$1 million per month, and keep their strategists, operations experts, and system developers busy for years’ (Davenport, 1996: p. 73). Finally the computing industry began to sell their software packages under the banner of BPR. For example, SAP, one of the biggest software vendors, now markets its software as a form of BPR.

Whether the idea of BPR will continue to thrive is not really the issue here. Certainly, the idea that firms needed to consider their business processes was an important one, which perhaps unfortunately got lost under the spot-light of ‘radical re-engineering’. The important point which this paper highlights is that the diffusion of this idea, regardless of its technical merits (or demerits), can be explained by considering the joint impact of micro-level organisational factors and meso- and macro-level contextual factors. The amount and type of ideas available to an organisation will be influenced by the boundary-spanning activities of individual organisational members and organisational characteristics, like size and strategy, will influence whether or how far these ideas are actually implemented. But the selection of ideas by individual boundary spanners does not occur in a vacuum—ideas do not simply float freely in time and space, ready to get selected when a particular individual suddenly recognises their potential either to solve an organisational problem or open up an opportunity. Meso- and macro-level variables which influence inter-organisational networking need to be recognised as important filters influencing who has access to ideas and what these ideas are. Thus, this paper expands the micro-level focus of innovation research by highlighting the importance of contextual factors at both the meso-industry level and the macro-national level which can also influence the innovation process.

However, while the results presented in this paper demonstrate the need to consider a combination of organisational, industry and national variables, it fails to fully consider their interaction. Because of the quantitative methodology, a factors or entitative approach which attempts to identify static forces which lead to the adoption of new technologies has been used (Hosking and Morley, 1992). Such an approach fails to fully exploit the dynamic, interactive and contextually-sensitive nature of innovation processes whereby new ideas are created, articulated, reinvented and manifested as particular new institutional and organisational forms and practices. What is needed now is a process approach which focuses on the dynamics of implementation, examining the behaviors of stakeholders over time. To develop a truly interactive perspective requires a more process-oriented methodology, for example using longitudinal case-studies to compare innovation processes in similar companies across different countries. This is a challenge to those who wish to extend our understanding of the innovation process.

In conclusion, successfully deploying BPR ideas is obviously very difficult, as evidenced by the estimated 70% failure rate (Hammer and Champy, 1993). Understanding the diffusion process more clearly can provide significant insights into the reasons for such a high degree of failure. Here, it has been shown how a combination of factors influences the diffusion process and, more importantly, how these factors can create the spread of fashions which inhibit the organisational learning necessary to effect the strategic changes associated with successful BPR. These insights can alert those involved in BPR projects, which will include many from the IS community, more fully to exploit the networks, both inter- and intra-organisational, which are the conduits through which the necessary knowledge owned by people working inside and outside the organisation can be absorbed and embedded in the new processes.

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